

MoMo SMS System

Purpose

The purpose of this testing was to confirm that the implemented database schema works correctly by inserting test data and verifying that database constraints are enforced.

1. Database Verification

Before inserting data, the database schema was verified using **SHOW TABLES** to ensure that all inputs would match the actual table structures.

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| momo_test |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> USE momo_test;
Database changed
mysql> SOURCE database/database_setup.sql;
Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_momo_test |
+-----+
| SystemLogs |
| TransactionCategories |
| Users |
+-----+
3 rows in set (0.00 sec)

mysql>
```

2. Users Table

Inserting valid records into the **Users** table.

```
mysql> INSERT INTO Users (userId, fullName, phoneNumber, userType) VALUES
  -> ('u1', 'Ahmad Daib', '+250795585810', 'PERSON'),
  -> ('u2', 'Tokiniaina Disaine', '+250752355833', 'PERSON'),
  -> ('u3', 'MTN MoMo Platform', NULL, 'BUSINESS');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Users;
+-----+-----+-----+-----+-----+
| userId | fullName      | phoneNumber | userType | createdAt      |
+-----+-----+-----+-----+-----+
| u1     | Ahmad Daib    | +250795585810 | PERSON  | 2026-01-24 22:41:20 |
| u2     | Tokiniaina Disaine | +250752355833 | PERSON  | 2026-01-24 22:41:20 |
| u3     | MTN MoMo Platform | NULL        | BUSINESS | 2026-01-24 22:41:20 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

3. SystemLogs Table

Inserting valid SMS log records into the **SystemLogs** table.

```
ahmad@Pavilion14: ~/Momo - x + v
mysql> DESCRIBE SystemLogs;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| logId      | char(36)      | NO   | PRI | NULL    |       |
| smsAddress | varchar(50)   | YES  |     | NULL    |       |
| messageBody | text          | NO   |     | NULL    |       |
| receivedAt | datetime      | NO   |     | NULL    |       |
| status     | enum('PENDING','PROCESSED','ERROR') | YES  |     | PROCESSED |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.09 sec)

mysql> INSERT INTO SystemLogs (logId, smsAddress, messageBody, receivedAt, status) VALUES
  -> ('l1', 'MTN-MoMo', 'You have received 5,000 RWF from Tokiniaina Disaine', '2026-01-24 10:15:00', 'PENDING'),
  -> ('l2', 'MTN-MoMo', 'You have paid 2,000 RWF to MTN MoMo Platform', '2026-01-24 11:00:00', 'PROCESSED');
Query OK, 2 rows affected (0.09 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM SystemLogs;
+-----+-----+-----+-----+-----+
| logId | smsAddress | messageBody | receivedAt | status |
+-----+-----+-----+-----+-----+
| l1    | MTN-MoMo  | You have received 5,000 RWF from Tokiniaina Disaine | 2026-01-24 10:15:00 | PENDING |
| l2    | MTN-MoMo  | You have paid 2,000 RWF to MTN MoMo Platform | 2026-01-24 11:00:00 | PROCESSED |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```

This confirms that SMS data is stored correctly.

4. TransactionCategories Table

The right transaction categories were inserted using the existing table structure.

```
mysql> INSERT INTO TransactionCategories (categoryId, categoryName) VALUES
-> ('c1', 'Peer Transfer'),
-> ('c2', 'Utility Payment');
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM TransactionCategories;
+-----+-----+-----+-----+
| categoryId | categoryName | categoryGroup | createdAt |
+-----+-----+-----+-----+
| c1         | Peer Transfer | UTILITY       | 2026-01-25 17:53:32 |
| c2         | Utility Payment | UTILITY       | 2026-01-25 17:53:32 |
+-----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql> |
```

This confirms correct category storage.

5. ENUM Constraint

An invalid **userType** value was inserted to test ENUM enforcement.

```
mysql> INSERT INTO Users (userId, fullName, userType)
-> VALUES ('u4', 'Invalid User', 'ADMIN');
ERROR 1265 (01000): Data truncated for column 'userType' at row 1
mysql> |
```

Result: The database rejected the invalid value.

ENUM values were also checked beforehand.

```
mysql> SHOW COLUMNS FROM Users LIKE 'userType';
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| userType | enum('PERSON','BUSINESS') | YES | | PERSON | |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.03 sec)

mysql> |
```

6. Primary Key Enforcement

A duplicate primary key was inserted into TransactionCategories.

```
mysql> INSERT INTO TransactionCategories (categoryId, categoryName)
-> VALUES ('c1', 'Duplicate Category');
ERROR 1062 (23000): Duplicate entry 'c1' for key 'TransactionCategories.PRIMARY'
mysql> |
```

Result: The database prevented duplicate records.

Conclusion

The database correctly enforces data integrity and prevents invalid data insertion. The successful test cases and the ones that failed prove that the database works as expected.